

FPGA LAB-Assignment-5

Design and Analysis of FIR Filters using Pipeline Architectures and IP Cores

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1 Objective

- To design and implement FIR filters using Verilog in both non-pipelined and pipelined architectures.
- To analyze and compare different FIR structures.
- To evaluate resource utilization for different implementations.
- To compare user-defined FIR filters with FIR IP cores.

2 Software Used

- Intel Quartus Prime Lite
- iverilog
- Python (for verification)

3 Theory

3.1 Finite Impulse Response (FIR) Filter

An FIR filter is a digital filter whose output depends on the present and past input samples. It is described as:

$$y[n] = \sum_{k=0}^{N-1} h[k] \cdot x[n-k]$$

where $h[k]$ represents filter coefficients and $x[n]$ is the input signal.

3.2 FIR Filter Architectures

3.2.1 Direct Form

In this structure, delayed input samples are multiplied with coefficients and summed to produce the output.

3.2.2 Transposed Form

The transposed form rearranges the structure to reduce delay and improve speed by pushing additions forward.

3.2.3 Symmetric (Optimized) Form

If coefficients are symmetric, computations can be reduced using pre-addition:

$$x[n - k] + x[n - (N - 1 - k)]$$

This reduces the number of multipliers by half.

3.3 Pipelining

Pipelining improves system speed by dividing computations into stages separated by registers. It increases throughput but introduces latency.

3.4 FIR IP Core

FIR IP cores are optimized hardware modules provided by FPGA vendors. They provide high performance and efficient resource utilization compared to manual designs.

4 Methodology

4.1 Non-Pipelined Design

Three FIR architectures were implemented:

- Direct Form
- Transposed Form
- Symmetric Form

4.2 Pipelined Design

The same architectures were modified using pipelining to improve performance by inserting registers in critical paths.

4.3 Hardware Synthesis

The designs were synthesized using Quartus Prime to analyze resource utilization.

5 Results

5.1 Simulation Results

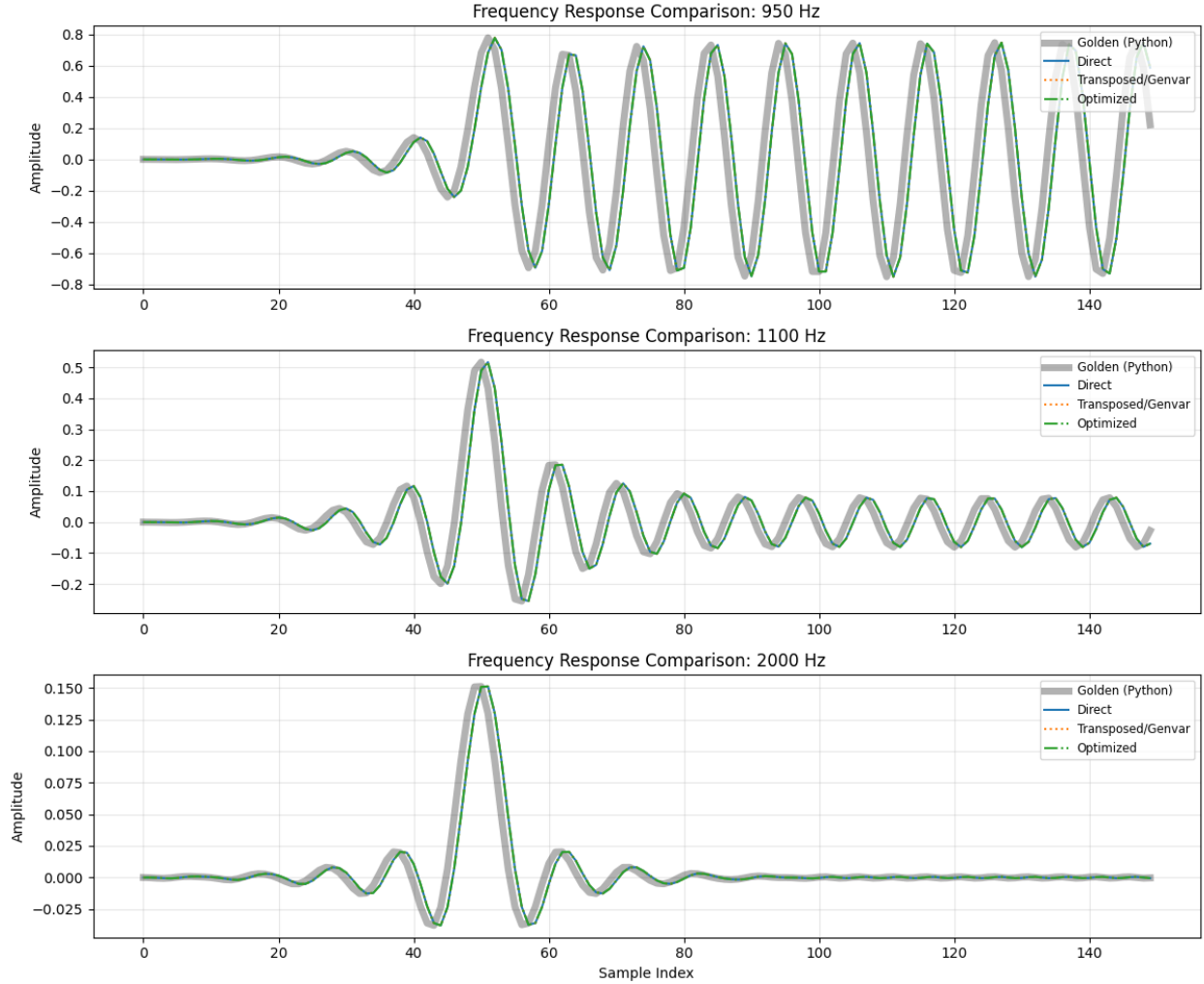


Figure 1: Pipeline Outputs

5.2 Observation

- The filter allows frequencies below cutoff to pass.
- Frequencies above cutoff are attenuated.
- Pipelined designs show correct outputs with added latency.

6 Resource Utilization

6.1 Direct Method

6.1.1 Non-pipelined

Flow Status	Successful - Wed Mar 25 13:01:19 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	FPGA-5
Top-level Entity Name	fir_direct
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	2,349 / 56,480 (4 %)
Total registers	1600
Total pins	34 / 268 (13 %)
Total virtual pins	1,600
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	99 / 156 (63 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 2: Resource Utilization

	Fmax	Restricted Fmax	Clock Name	Note
1	52.45 MHz	52.45 MHz	clk	

Figure 3: F_{max}

6.1.2 Pipelined

Flow Status	Successful - Wed Mar 25 22:19:30 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	FPGA-5
Top-level Entity Name	fir_direct_pipeline
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	2,486 / 56,480 (4 %)
Total registers	2496
Total pins	66 / 268 (25 %)
Total virtual pins	1,600
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	99 / 156 (63 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 4: Resource Utilization

	Fmax	Restricted Fmax	Clock Name	Note
1	59.58 MHz	59.58 MHz	clk	

Figure 5: F_{max}

6.2 Genvar Method

6.2.1 Non-pipelined

Flow Status	Successful - Wed Mar 25 12:55:00 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	FPGA-5
Top-level Entity Name	fir_genvar
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	2,349 / 56,480 (4 %)
Total registers	1600
Total pins	34 / 268 (13 %)
Total virtual pins	1,600
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	99 / 156 (63 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 6: Resource Utilization

	Fmax	Restricted Fmax	Clock Name	Note
1	52.45 MHz	52.45 MHz	clk	

Figure 7: F_{max}

6.2.2 Pipelined

Flow Status	Successful - Wed Mar 25 22:57:14 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	FPGA-5
Top-level Entity Name	fir_transposed_sym
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	2,915 / 56,480 (5 %)
Total registers	4769
Total pins	66 / 268 (25 %)
Total virtual pins	800
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	50 / 156 (32 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 8: Resource Utilization

	Fmax	Restricted Fmax	Clock Name	Note
1	216.5 MHz	216.5 MHz	clk	

Figure 9: F_{max}

6.3 Optimized Method

6.3.1 Non-pipelined

Flow Status	Successful - Wed Mar 25 12:47:50 2026		
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition		
Revision Name	FPGA-5		
Top-level Entity Name	fir_optimized		
Family	Cyclone V		
Device	5CGXFC7C7F23C8		
Timing Models	Final		
Logic utilization (in ALMs)	1,976 / 56,480 (3 %)		
Total registers	1600	1,976 / 56,480 (3 %)	
Total pins	34 / 268 (13 %)		
Total virtual pins	1,600		
Total block memory bits	0 / 7,024,640 (0 %)		
Total DSP Blocks	49 / 156 (31 %)		
Total HSSI RX PCSs	0 / 6 (0 %)		
Total HSSI PMA RX Deserializers	0 / 6 (0 %)		
Total HSSI TX PCSs	0 / 6 (0 %)		
Total HSSI PMA TX Serializers	0 / 6 (0 %)		
Total PLLs	0 / 13 (0 %)		
Total DLLs	0 / 4 (0 %)		

Figure 10: Resource Utilization

	Fmax	Restricted Fmax	Clock Name	Note
1	52.07 MHz	52.07 MHz	clk	

Figure 11: F_{max}

6.3.2 Pipelined

Flow Status	Successful - Wed Mar 25 22:46:34 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	FPGA-5
Top-level Entity Name	fir_optimized1
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	12,323 / 56,480 (22 %)
Total registers	46092
Total pins	66 / 268 (25 %)
Total virtual pins	800
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	50 / 156 (32 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 12: Resource Utilization

	Fmax	Restricted Fmax	Clock Name	Note
1	115.29 MHz	115.29 MHz	clk	

Figure 13: F_{max}

7 FIR IP Core Implementation

7.1 Design using FIR IP

The FIR filter was implemented using the FIR IP core available in Quartus.

7.2 Resource Utilization

Flow Status	Successful - Wed Mar 25 18:50:06 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	FPGA-5
Top-level Entity Name	fir_ip_core
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	11 / 56,480 (< 1 %)
Total registers	40
Total pins	46 / 268 (17 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	1 / 156 (< 1 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 14: Resource Util

	Fmax	Restricted Fmax	Clock Name	Note
1	300.03 MHz	220.02 MHz	clk	lim...in)

Figure 15: F_{max}

7.3 Comparison with User Design

Table 1: Resource Utilization Comparison of FIR Filter Implementations

Implementation	Logic Utilization (ALMs)	Total Registers	DSP Blocks
Direct Form (<code>fir_direct</code>)	2,349 (4%)	1600	99 (63%)
Transposed (<code>fir_genvar</code>)	2,349 (4%)	1600	99 (63%)
Optimized Symmetric (<code>fir_optimized</code>)	1,976 (3%)	1600	49 (31%)
FIR Compiler IP (<code>fir_ip_core</code>)	11 ($\approx 1\%$)	40	1 ($\approx 1\%$)

8 Discussion

- Pipelining increases speed but also increases latency.
- Symmetric FIR reduces hardware complexity.
- FIR IP cores provide optimized performance and lower resource usage.

9 Conclusion

- FIR filters were successfully implemented using different architectures.
- Pipelining significantly improved performance.
- FIR IP cores outperformed manual implementations in terms of efficiency.